JAN-W-472

29 MAY 1947

JOINT ARMY-NAVY SPECIFICATION

WIRE, LOW-BRASS (ARMING WIRE)

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and supersedes the following specification:

U. S. Army — Navy Department
57-217A
12 May 1939

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATION

A-1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

U. S. ARMY SPECIFICATIONS

94-40645—Marking; Shipment. (Domestic and Export).[1]
100-2—Standard Specification for Marking Shipments by Contractors.[2]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[3]

FEDERAL SPECIFICATIONS

NN-B-591—Boxes; Wood-Cleated-Fiberboard.
NN-B-601—Boxes; Wood-Cleated-Plywood (For Domestic Shipment).
NN-B-621—Boxes; Wood, Nailed and Lock-Corner.
NN-B-631—Boxes; Wood, Wirebound, For Domestic Shipment.
LLL-B-631—Boxes; Fiber, Corrugated (For Domestic Shipment).
LLL-B-636—Boxes; Fiber, Solid (For Domestic Shipment).

A-2. *Other publication.*—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[3]

B. GRADE

B-1. This specification covers one grade of low-brass wire as hereinafter specified.

C. MATERIAL AND WORKMANSHIP

C-1. *Material.*—Low-brass wire shall be made from high-grade copper and high-grade zinc or from high-grade ingot metals

[1] Applicable only to Army Air Forces purchases.
[2] Applicable only to Army purchases, except Army Air Forces.
[3] Applicable only to Navy purchases.

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and clean scrap of known composition, which will produce material meeting the requirements of this specification.

C–2. *Workmanship.*—Low-brass wire shall be uniform in composition and quality, and free from slivers, kinks, laps, seams, scratches, indentations, and other injurious defects. It shall be new, bright, smooth, and of uniform size.

D. GENERAL REQUIREMENTS

D–1. See section E.

E. DETAIL REQUIREMENTS

E–1. *Chemical composition.*—

E–1a. Low-brass wire shall meet the requirements shown in table I.

TABLE I.—*Chemical requirements.*

Percent composition			
Copper	Lead (max.)	Iron (max.)	Zinc
79.0 - 81.0	0.05	0.05	Remainder

E–1b. An analysis of each lot shall be furnished by the contractor, showing the percentage of each element designated in table I.

E–2. *Physical properties.*—

E–2a. *Tensile.*—Low-brass wire shall have the following tensile properties:

Tensile strength.—Not less than 100,000 p.s.i. nor more than 130,000 p.s.i.

Elongation in 10 inches.—Not less than 1 percent.

E–2b. *Wrapping.*—Low-brass wire shall withstand, without fracture, being wrapped at normal room temperature around its own diameter eight consecutive turns with a pitch substantially equal to the diameter of the wire.

E–3. *Permissible variations in diameter.*—The diameter of the wire shall not vary from the specified diameter by more than the amount shown in table II.

TABLE II.—*Permissible variations in diameter.*

Diameter	Permissible variations
Inch	*Inch*
0.025 to 0.040 incl.	±0.001
Over 0.040 to 0.100 incl.	±0.002

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F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Size of lot.*—Unless otherwise specified in the contract or order, a lot shall consist of all wire of the same size, submitted for inspection at the same time.

F–2. *Sampling.*—

F–2a. *Tension and wrapping-test samples.*—The inspector shall select six test specimens from each coil in the lot. Three pieces, each 20 inches long, shall be selected for tension tests. Three pieces, each 10 inches long, shall be selected for wrapping tests. Test specimens shall be free from kinks and bends other than the curvature caused by coiling.

F–2b. *Chemical test samples.*—The inspector shall select one sample for Government analysis from each 5,000 pounds or less in the lot. Each sample shall weigh not less than 2 ounces and, whenever practicable, shall consist of equal amounts taken from at least four coils. Samples may be taken from pieces used for the wrapping or the tension tests.

F–3. *Inspection.*—Unless otherwise specified, inspection shall be made at the point of delivery.

F–4. *Tests.*—Laboratory tests shall be made in accordance with paragraphs F–4a to F–4c inclusive. Unless otherwise specified, tests on Navy purchases shall be made at a Government or other approved laboratory.

F–4a. *Chemical analysis.*—Compliance with the requirements of paragraph E–1 shall be determined by approved analytical methods at the laboratory designated in the contract or order (see par. H–2) or by the inspector.

F–4b. *Tension tests.*—The test specimens shall be tested in a suitable testing machine, which shall be calibrated when required by the inspector. The average of the results of three tests from each coil shall determine acceptance or rejection of the coil. The elongation shall be determined as the permanent increase in length, due to the breaking of the wire in tension, measured between gage marks placed upon the wire originally 10 inches apart. The fracture shall be between the gage marks and not closer than 1 inch from either gage mark.

F–4c. *Wrapping tests.*—The test specimens selected for the wrapping tests shall meet the requirements prescribed in paragraph E–2b. Failure of one or more test specimens shall be cause for rejection of the coil represented.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packaging for domestic shipment.*—The wire shall be wound in coils weighing approximately 40 pounds each. Each coil shall be wrapped to protect the wire from damage. Inside diameter of coil shall be as follows:

Diameter of wire	Inside diameter of coil
Inch	*Inches*
0.025 to 0.036	5
0.037 to 0.048	9
0.049 to 0.100	12

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G–2. *Packing for domestic shipment.*—Wire, packaged as described in paragraph G–1, shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carrier, at the lowest rate to the point of delivery. When wood-cleated-fiberboard boxes, wood-cleated-plywood boxes, nailed wood boxes, wirebound wood boxes, corrugated fiber boxes, or solid fiber boxes are used, they shall conform to the requirements of Federal Specifications NN–B–591, NN–B–601, NN–B–621, NN–B–631, LLL–B–631, and LLL–B–636, respectively. The gross weight of wood boxes shall not exceed approximately 200 pounds; of fiberboard boxes, 70 pounds.

G–3. *Marking.*—

G–3a. *Coils.*—To each coil shall be securely attached a tag with reinforced eyelet, on which shall be typed or clearly and indelibly written, the name and size of the material, specification number, weight of coil, name, brand or trade mark of the manufacturer, and the contract or order number. The tags shall be shellacked on the information side after the information has been typed or written thereon.

G–3b. *Shipping containers.*—In addition to any special marking required by the contract or order, shipping containers for the Army, except Army Air Forces, shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for Army Air Forces in accordance with the requirements of U. S. Army Specification 94–40645; for the Navy in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES

H–1. *Use.*—The wire covered by this specification is for use primarily as arming wire for bombs.

H–2. Requests, requisitions, schedules, and contracts or orders should contain the following:

(a) Title, number and date of the specification.

(b) Place of testing for chemical composition. (See par. F–4a.)

H–3. Contractors supplying low-brass wire may be required to replace material in which injurious defects are revealed by manufacturing operations performed subsequent to acceptance, without expense to the Government.

H–4. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy, U. S. Army and Federal specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

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H–5. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–6. Copies of this Joint Army-Navy specification (required for Army purchases) may be obtained from the Office, Chief of Ordnance, War Department, Washington 25, D. C.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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